

CONCEPT MAP

SORTING

Arranges data in ascending/descending order

MERGE SORT

DIVIDE PHASE

CONQUER PHASE

COMBINE PHASE

RECURRENCE RELATION

$$T(n) = 2T(n/2) + O(n)$$

$2T(n/2)$ → Two recursive calls arrays
 $O(n)$ → Linear time for merging

TIME COMPLEXITY

$$O(n \log n)$$

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CONCEPT MAP

RECURRENCE RELATION

EXPANSION PATTERN

SUMMATION

COMPLEXITY

SUBSTITUTION METHOD

SOLVING RECURRENCES
EFFECTIVELY

Describes a problem in terms of smaller subproblems

Sum the series obtained from expansion to get a closed form.

Prove the guessed bound correct using induction

Expand the recurrence repeatedly to reveal a pattern

Determine the order of growth using Big-O notation.

Expansion shows the pattern

Summation gives the closed form

Substitution proves the bound

Together they yield the final complexity

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